

Jason D. Hartline

Professor of Computer Science
Northwestern University
Evanston, IL 60208.

hartline@northwestern.edu
<https://jasonhartline.com/>
+1 (415) 200-6171

Research Interests

 **Economics.** Mechanism design, microeconomics, econometrics, economics of algorithms.

 **Law.** Algorithmic law, accountability, regulating algorithms, science of law.

 **Artificial Intelligence.** Explainability, fairness, alignment, future of work.

 **Data Science.** Machine learning, outcomes from learning agents, regulating algorithms.

Education

Ph.D. in Computer Science. University of Washington, Seattle, WA. *Summer 2003*
Thesis: Optimization in the Private Value Model: Competitive Analysis Applied to Auction Design
Advisor: Anna Karlin.

M.S. in Computer Science. University of Washington, Seattle, WA. *Spring 2000*

B.S. in Computer Science. Cornell University, Ithaca, NY. *Spring 1997*

B.S. in Electrical Engineering. Cornell University, Ithaca, NY. *Spring 1997*

Current Appointment

Professor. Northwestern U., Evanston, IL. *Fall 2019 – present*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Cofounder. Virtual Chair Inc., Chicago IL. *Summer 2020 – present*

Previous Appointments

Visiting Professor. Stanford U., Stanford, CA. *2023-2024*
Economics department and Graduate School of Business

Associate Professor. Northwestern U., Evanston, IL. *Fall 2012 – Summer 2019*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Visiting Researcher. Microsoft Research, Cambridge, MA. *Spring 2015*

Visiting Professor. Harvard U., Cambridge, MA. *2014*
Computer Science and Economics departments.

Assistant Professor. Northwestern U., Evanston, IL. *Winter 2008 – Summer 2012*
 Electrical Engineering and Computer Science Department, McCormick School of Engineering and
 Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy).

Researcher. Microsoft Research, Mountain View, CA. *2004 – 2007*
 Research Area: Algorithmic Mechanism Design, Auction Theory, Pricing Algorithms, Auctions for
 Sponsored Search.

Post-doctoral Research Fellow. ALADDIN, Carnegie Mellon U., Pittsburgh, PA. *Fall 2003*
 Research Area: Mechanism Design.
 Supervisor: Avrim Blum.

Mentoring

Ph.D. Advisees. *current*
 Anant Shah, Matthew vonAllmen, Chang Wang, Yifan Wu, Chenhao Zhang.

Former Students. *since 2009*
 Modibo Camara (Stanford), Yingkai Li (Yale), Yiding Feng (Microsoft Research), Bach Ha (Mi-
 crosoft), Aleck Johnsen, Nima Haghpahan (Penn State, Asst. Prof.), Darrell Hoy (Carta), Michalis
 Mamakos (Northwestern, postdoc), and Samuel Taggart (Oberlin, Asst. Prof.)

Post-doctoral Fellows. *since 2006*
 Jinshuo Dong, Hedyeh Beyhaghi (CMU), Liad Blumrosen (Hebrew U.), Azarakhsh Malekian
 (Toronto)

Short-term Students. *since 2004*
 Gagan Aggarwal, Abraham Flaxman, Ning Chen, Mukund Sundararajan, Benjamin Prosnitz,
 Matthew Burgess, Saeed Alaei, Hu Fu, Shweta Jain, Rad Niazadeh, Sarah Lim, Sadie Hood.

Service

 **Co-organizer.** Monthly CS+Law Workshop *2021 – present*
<https://www.cslawworkshop.org/>

 **Program Committee.** ACM Conf. on Economics & Computation. *2005, 2006, 2008 – present*

  **Program Committee.** Symposium on Foundations of Responsible Computing *2024*

 **Program Chair.** ACM Conference on Economics and Computation. *2023*

 **Program Committee.** ACM Symposium on CS and Law. *2023*

 **Associate Editor.** Journal of Economic Theory *2019 – present*

  **Co-organizer.** IDEAL Special Quarter on Data Economics. *Fall 2022*

 **Co-organizer.** IDEAL Special Quarter on CS and Law. *Spring 2021*

 **Virtual Chair.** ACM Conference on Economics and Computation *2020*

 **Co-organizer.** Northwestern CS+Econ Quarterly Workshop *2018 – 2020*

Co-organizer. Northwestern Quarterly Theory Workshop *2016 – 2020*

 **General Chair.** ACM Conference on Electronic Commerce. *2020*

🏠	🏠 Co-organizer. Special Quarter on Online Markets and Data Science with Jacob Abernethy, Constantinos Daskalakis, and Denis Nekipelov.	<i>Spring 2018</i>
🏠	Special Initiatives Chair. ACM Special Interest Group on E-commerce. on the Academic Job Market.	<i>2014 – 2015</i>
🏠	Guest Editor. Games and Economic Behavior. special issues for papers from STOC, FOCS, and SODA conferences.	<i>2011 – 2014</i>
🏠	Advisory Editor. Games and Economic Behavior.	<i>2012 – 2017</i>
	Associate Editor. Operations Research Letters.	<i>2011 – 2014</i>
🏠	Co-organizer. New York Computer science and Economics (NYCE) Day.	<i>2013</i>
🏠	Co-organizer. FOCS Workshop on Bayesian Mechanism Design.	<i>2014</i>
	Program Committee. Symposium on Theory of Computation	<i>2012</i>
🏠	Co-organizer. Workshop on Bayesian Mechanism Design	<i>2011</i>
🏠	Co-organizer. Greece Economic and Algorithmic Theory Week.	<i>2011, 2014</i>
🏠	Co-organizer. Bertinoro Workshop on Algorithmic Game Theory.	<i>2006, 2010</i>
🏠	Tutorials Chair. ACM Conference on Electronic Commerce.	<i>2010</i>
🏠	Local Arrangements. ACM Conference on Electronic Commerce.	<i>2008</i>
	Organizer. Midwest Theory Day.	<i>2008</i>
	Program Committee. ACM-SIAM Symposium on Discrete Algorithms.	<i>2007</i>
🏠	Co-organizer. Bay Algorithmic Game Theory Symposium (biannual).	<i>2006 – 2007</i>
🏠	Co-organizer. Workshop on Sponsored Search Auctions.	<i>2006</i>
🏠	Co-organizer. Alternative Solution Concepts in Mechanism Design.	<i>2006</i>
🏠	Co-organizer. ALADDIN Workshop on Auction Theory & Practice.	<i>2003</i>

Awards, Fellowships, and Grants

🏠	🏠 FORC Best Student Paper. with Jiale Chen and Onno Zoeter for “Fair Grading of Randomized Exams”	<i>2023</i>
🏠	NSF AF. Mechanism Design for the Classroom	<i>2022</i>
🏠	ESA Test of Time. with Andrew Goldberg for “Competitive Auctions and Multiple Digital Goods” from ESA 2001.	<i>2021</i>
🏠	SIGecom Test of Time. with Andrew Goldberg and Andrew Wright for “Competitive Auctions and Digital Goods” from SODA 2021.	<i>2021</i>
🏠	🏠 NSF TRIPODS. Institute for Data, Econometrics, Algorithms and Learning co-dorector; with Aravindan Vijayaraghavan and others.	<i>2019</i>

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NSF AitF. Mechanism Design and Machine Learning for Peer Grading
 with Douglas Downey and Eleanor O'Rourke. 2017
- 
NSF AF. Non-revelation Mechanism Design 2016
- Teacher of the Year.** EECS Dept., Northwestern U. 2011
- 

NSF ICES. Towards Realistic Mechanisms: statistics, inference, and approximation in simple Bayes-Nash implementation. With Shuchi Chawla and Denis Nekipelov. 2011
- 
NSF CAREER Award. Mechanism Design 2009
- 
NSF TF. Mechanism Design and Approximation
 with Shuchi Chawla. 2008
- ALADDIN Post-doctoral Research Fellowship.** Carnegie Mellon University. 2003
- NSF Math Sciences Post-doctoral Research Fellowship.** Declined. 2003
- Microsoft Endowed Fellowship.** CS Dept., U. of Washington 2001
- Bob Bandes Teaching Award, Honorable Mention.** CS Dept., U. of Washington 1998
- Small Business Innovative Research Grant.** Department of Education 1997

Patents

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Online Pricing and Buyback. U.S. Patent #8260724 2012
 with Moshe Babaioff and Robert Kleinberg.
- 
Systems and Methods for Pricing and Selling Digital Goods. U.S. Patent #6985885 2006
 with Andrew Goldberg and Andrew Wright.

Book Chapters

- 
Profit Maximizing Mechanism Design. Algorithmic Game Theory 2007
 with Anna Karlin; eds. Noam Nisan, Tim Roughgarden, Eva Tardos, and Vijay Vazirani.

Popular Press

- 
Badminton and the Science of Rule Making. *Huffington Post* 2012
 with Robert Kleinberg.

Working Papers

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ElicitationGPT: Text Elicitation Mechanisms via Language Models. 2024
 with Yifan Wu.
- Subgame Optimal and Prior-Independent Online Algorithms.** 2024
 with Aleck Johnsen and Anant Shah.
- 

Algorithmic robust forecast aggregation. 2024
 with Yongkang Guo, Zhihuan Huang, Yuqing Kong, Anant Shah, and Fang-Yi Yu.

- 🏠 **Mechanism Redesign.** 2017
with Shuchi Chawla and Denis Nekipelov.

Journal Papers

- 🏠 **Fast Core Pricing for Rich Advertising Auctions.** *Operations Research* 2022
with Rad Niazadeh, Nicole Immorlica, Mohammad Resa Khani, and Brendan Lucier.
- 🏠 **When is pure bundling optimal?.** with Nima Haghpahanah *Review of Economic Studies* 2021
- 🏠 **Bernoulli factories and black-box reductions in mechanism design.** *JACM* 2021
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh.
- 🏠 **Efficient Computation of Optimal Auctions via Reduced Forms.** *Math of OR* 2019
with Saeed Alaei, Hu Fu, Nima Haghpahanah, and Azarakhsh Malekian.
- 🏠 **Optimal auctions vs. Anonymous Pricing.** *Games and Economic Behavior* 2018
with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan. Special issue.
- 🏠 **Non-optimal Mechanism Design.** *American Economic Review* 2015
with Brendan Lucier.
- 🏠 **Bayesian Incentive Compatibility and Matchings.** *Games and Economic Behavior* 2015
with Robert Kleinberg and Azarakhsh Malekian. Special issue.
- 🏠 **Mechanism Design via Consensus Estimates, Cross Checking, and Profit Extraction.**
with Bach Ha. Special issue. *Transactions on Economics and Computation* 2013
- 🏠 **Optimal Crowdsourcing Contests.** *Games and Economic Behavior* 2015
with Shuchi Chawla and Balu Sivan. Special issue.
- 🏠 **Envy freedom and prior-free mechanism design.** *Journal of Economic Theory* 2015
with Nikhil Devanur and Qiqi Yan. Special issue.
- 🏠 **Bayesian Mechanism Design.** *FTTCS*¹ 2012
- 🏠 **Approximation in Mechanism Design.** *American Economic Review* 2012
- 🏠 **Derandomization of Auctions.** *Games and Economic Behavior* 2010
with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.
- 🏠 **Algorithms for Data Migration.** *Algorithmica* 2010
with Eric Anderson, Joseph Hall, M. Hobbess, Anna Karlin, Jared Saia, Ram Swaminathan, and John Wilkes.
- 🏠 **Reducing Mechanism Design to Algorithm Design via Machine Learning.** *JCSS*² 2008
with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.
- 🏠 **Competitive Auctions.** *Games and Economic Behavior* 2006
with Andrew Goldberg, Anna Karlin, Mike Saks, and Andrew Wright. Special issue.
- 🏠 **Characterizing History Independent Data Structures.** *Algorithmica* 2005
with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke. Special issue.

¹Foundations and Trends in Theoretical Computer Science.

²Journal of Computer and Systems Sciences.

Refereed Conference Papers

- 
Fundamental Limits of Throughput and Availability: Applications to prophet inequalities and transaction fee mechanism design. *EC*⁸ 2024
 with Adityan Ganesh, Atanu Sinha, and Matthew vonAllmen
- 
A Decision Theoretic Framework for Measuring AI Reliance. *FAccT*⁴ 2024
 with Ziyang Guo, Yifan Wu, and Jessica Hullman.
- 

Designing Shared Info Displays for Agents of Varying Strategic Sophist. *CSCW*⁵ 2024
 with Dongping Zhang and Jessica Hullman.
- 

Error-Tolerant E-Discovery Protocols. *CSLaw*⁶ 2024
 with Jinshuo Dong, Liren Shan, and Aravindan Vijayaraghavan.
- 


Regulation of algorithmic collusion. with Sheng Long and Chenhao Zhang *CSLaw* 2024
- 
Equivocal Blends: Prior Independent Lower Bounds. *ITCS*⁷ 2024
 with Aleck Johnsen
- 
The Rational Agent Benchmark for Data Visualization. *VIS*⁸ 2023
 with Yifan Wu, Ziyang Guo, Michalis Mamakos, Jessica Hullman
- 
Optimal Scoring Rules for Multi-dimensional Effort. *COLT*⁹ 2023
 with Liren Shan, Yingkai Li, and Yifan Wu
- 

Fair Grading Algorithms for Randomized Exams. *FORC*¹⁰ 2023
 with Jiale Chen and Onno Zoeter. Best student paper award.
- 

Screening with Disadvantaged Agents. *FORC* 2023
 with Hedyeh Beyhaghi, Modibo Camara, Aleck Johnsen, and Sheng Long
- 


Non-strategic Econometrics (for Initial Play). *AAMAS*¹¹ 2023
 with Daniel Chui and James Wright
- 
Simple Mechanisms for Non-linear Agents. *SODA*¹² 2023
 with Yiding Feng and Yingkai Li
- 

Algorithmic Learning Foundations for Common Law. *CSLaw* 2022
 with Dan Linna, Liren Shan, and Alex Tang
- 


Classification Protocols with Minimal Disclosure. *CSLaw* 2022
 with Jinshuo Dong and Aravindan Vijayaraghavan

³ACM Conference on Economics and Computation

⁴ACM Conference on Fairness, Accountability, and Transparency

⁵ACM Conference on Computer-Supported Cooperative Work and Social Computing

⁶ACM Symposium on Computer Science and Law

⁷Innovations in Theoretical Computer Science Conference

⁸IEEE Transactions on Visualization and Computer Graphics

⁹Conference on Learning Theory.

¹⁰Symposium on Foundations of Responsible Computing.

¹¹International Conference on Autonomous Agents and Multiagent Systems

¹²ACM-SIAM Symposium on Discrete Algorithms.

- Karp: A Language for NP Reductions.** *PLDI*¹³ 2022
with Chenhao Zhang and Christos Dimoulas
- Visualization Equilibrium.** *IEEE VIS*¹⁴ 2021
with Paula Kayongo, Glen Sun, and Jessica Hullman
- Revelation Gap for Pricing from Samples.** with Yiding Feng and Yingkai Li *STOC*¹⁵ 2021
- Welfare-maximizing Guaranteed Dashboard Mechanisms.** *EC* 2021
with Yuan Deng, Jieming Mao, and Balasubramanian Sivan
- Non-quasi-linear Agents in Quasi-linear Mechanisms.** *ITCS* 2021
with Moshe Babaioff, Richard Cole, Nicole Immorlica, and Brendan Lucier
- Mechanisms for a no-regret agent: Beyond the common prior.** *FOCS*¹⁶ 2020
with Modibo Camara and Aleck Johnsen
- Benchmark design and prior-independent optimization.** *FOCS* 2020
with Aleck Johnsen and Yingkai Li
- Inference from Prices.** *SODA* 2020
with Aleck Johnsen, Denis Nekipelov, and Zihe Wang.
- A Truthful Cardinal Mechanism for One-Sided Matching.** *SODA* 2020
with Rediet Abebe, Richard Cole, and Vasilis Gkatzelis.
- Sample Complexity for Non-Truthful Mechanisms.** *EC* 2019
with Samuel Taggart.
- Dashboard Mechanisms for Online Marketplaces.** *EC* 2019
with Aleck Johnsen, Denis Nekipelov, and Onno Zoeter.
- Optimal Auctions vs. Anonymous Pricing: Beyond Linear Utility.** *EC* 2019
with Yiding Feng and Yingkai Li.
- An End-to-end Argument in Mechanism Design (Prior-independent Auctions for Budgeted Agents).** With Yiding Feng. *FOCS* 2018
- Fast Core Pricing for Rich Advertising Auctions.** *EC* 2018
with Nicole Immorlica, Mohammad Reza Khani, Brendan Lucier, and Rad Niazadeh.
- Bernoulli Factories and Black-box Reductions in Mechanism Design.** *STOC* 2017
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh
- Bayesian Budget Feasibility with Posted Pricing.** *WWW*¹⁷ 2016
with Eric Balkanski.
- A/B Testing of Auctions.** *EC* 2016
with Shuchi Chawla and Denis Nekipelov.

¹³ACM SIGPLAN International Conference on Programming Language Design and Implementation¹⁴IEEE Transactions on Visualization and Computer Graphics¹⁵ACM Symposium on Theory of Computing.¹⁶IEEE Symposium on Foundations of Computer Science.¹⁷International Conference on the World Wide Web.

- No-regret Learning in Bayesian Games.** *NeurIPS*¹⁸ 2015
 with Vasilis Syrgkanis and Eva Tardos.
- Reverse Mechanism Design.** *EC* 2015
 with Nima Haghpanah.
- Optimal auctions vs. Anonymous Pricing.** *FOCS* 2015
 with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan.
- Mechanism Design for Data Science.** *EC* 2014
 with Shuchi Chawla and Denis Nekipelov.
- Price of Anarchy for Auction Revenue.** *EC* 2014
 with Darrell Hoy and Samuel Taggart.
- Optimal Auctions for Correlated Buyers with Sampling.** *EC* 2014
 with Hu Fu, Nima Haghpanah, and Robert Kleinberg.
- The Simple Economics of Approximately Optimal Auctions.** *FOCS* 2013
 with Saeed Alaei, Hu Fu, and Nima Haghpanah.
- Auctions with Unique Equilibria.** *EC* 2013
 with Shuchi Chawla.
- Prior-independent Auctions for Risk-averse Agents.** *EC* 2013
 with Hu Fu and Darrell Hoy.
- Prior-free Auctions for Budgeted Agents.** *EC* 2013
 with Nikhil Devanur and Bach Ha.
- Prior-independent Mechanisms for Scheduling.** *STOC* 2013
 with Shuchi Chawla, David Malec, and Balu Sivan.
- Mechanism Design via Multi- to Single-agent Reduction.** *EC* 2012
 with Saeed Alaei, Hu Fu, Nima Haghpanah, and Azarakhsh Malekian.
- Optimal Crowdsourcing Contests.** *SODA* 2012
 with Shuchi Chawla and Balu Sivan. Invited to GEB special issue.
- Mechanism Design via Consensus Estimates, and Cross Checking, and Profit Extraction.** *SODA* 2012
 With Bach Ha. Invited to TEAC special issue.
- Truth, Envy, and Profit.** *EC* 2011
 with Qiqi Yan. Invited to JET special issue.
- Bayesian Incentive Compatibility and Matchings.** *SODA* 2011
 with Robert Kleinberg and Azarakhsh Malekian. Invited to GEB special issue.
- Bayesian Algorithmic Mechanism Design.** *STOC* 2010
 with Brendan Lucier.
- Sequential Posted Pricing and Multi-parameter Mechanism Design.** *STOC* 2010
 with Shuchi Chawla, David Malec, and Balasubramanian Sivan.

¹⁸Conference on Neural Information Processing Systems

- Simple versus Optimal Mechanisms.** *EC 2009*
 with Tim Roughgarden.
- Limited and Online Supply and the Bayesian Foundations of Prior-free Mechanism Design.** *EC 2009*
 with Nikhil Devanur.
- Selling Ad Campaigns: Online Algorithms with Cancellations.** *EC 2009*
 with Moshe Babaioff and Robert Kleinberg.
- Mechanism Design and Money Burning.** *STOC 2008*
 with Tim Roughgarden.
- Optimal Marketing Strategies over Social Networks.** *WWW 2008*
 with Vahab Mirrokni and Mukund Sundararajan.
- Auctions for Structured Procurement.** *SODA 2008*
 with Matthew Cary, Abraham Flaxman, and Anna Karlin.
- Algorithmic Pricing via Virtual Valuations.** *EC 2007*
 with Shuchi Chawla and Robert Kleinberg.
- Knapsack Auctions.** *2006*
 with Gagan Aggarwal.
- Bayesian Optimal No-deficit Mechanism Design.** *WINE¹⁹ 2006*
 with Shuchi Chawla, R. Ravi, and Uday Rajan.
- Mechanism Design via Machine Learning.** *FOCS 2005*
 with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.
- Derandomization of Auctions.** *STOC 2005*
 with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.
- On Profit-Maximizing Envy-Free Pricing.** *SODA 2005*
 with Venkat Guruswami, Anna Karlin, David Kempe, Claire Kenyon, and Frank McSherry.
- Collusion-Resistant Mechanisms for Single Parameter Agents.** *SODA 2005*
 with Andrew Goldberg.
- Near-Optimal Online Auctions.** *SODA 2005*
 with Avrim Blum.
- From Optimal Limited to Unlimited Supply Auctions.** *EC 2005*
 with Robert McGrew.
- On the Competitive Ratio of the Random Sampling Auction.** *WINE 2005*
 with Uriel Feige, Abraham Flaxman, and Robert Kleinberg.
- Near-Optimal Pricing in Near-Linear Time.** *WADS²⁰ 2005*
 with Vladlen Koltun.

¹⁹Conference on Web and Internet Economics.

²⁰Workshop on Algorithms and Data Structures.

- A Lower Bound on the Competitive Ratio of Truthful Auctions.** *STACS*²¹ 2004
 with Andrew Goldberg, Anna Karlin, and Mike Saks.
- Competitiveness via Consensus.** *SODA* 2003
 with Andrew Goldberg
- Envy-Free Auctions for Digital Goods.** *EC* 2003
 with Andrew Goldberg.
- Truthful and Competitive Double Auctions.** *ESA*²² 2002
 with Kaustubh Deshmukh, Andrew Goldberg, and Anna Karlin.
- Competitive Generalized Auctions.** *STOC* 2002
 with Amos Fiat, Andrew Goldberg, and Anna Karlin.
- Characterizing History Independent Data Structures.** *ISAAC*²³ 2002
 with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke.
- Competitive Auctions for Multiple Digital Goods.** *ESA* 2001
 with Andrew Goldberg.
- Competitive Auctions and Digital Goods.** *SODA* 2001
 with Andrew Goldberg and Andrew Wright.
- On Algorithms for Efficient Data Migration.** *SODA* 2001
 with Joe Hall, Anna Karlin, Jared Saia, and John Wilkes.
- An Experimental Study of Data Migration Algorithms.** *WAE*²⁴ 2001
 with E. Anderson, J. Hall, M. Hobbes, A. Karlin, J. Saia, R. Swaminathan, and J. Wilkes.

²¹Symposium on Theoretical Aspects of Computer Science.

²²European Symposium on Algorithms.

²³International Symposium on Algorithms and Computation.

²⁴International Workshop on Algorithm Engineering.